Efficacy of Ajwa Date Seed Powder on Human Lipid Profile: A Randomized, Double-blind, Placebo-controlled Clinical Trial

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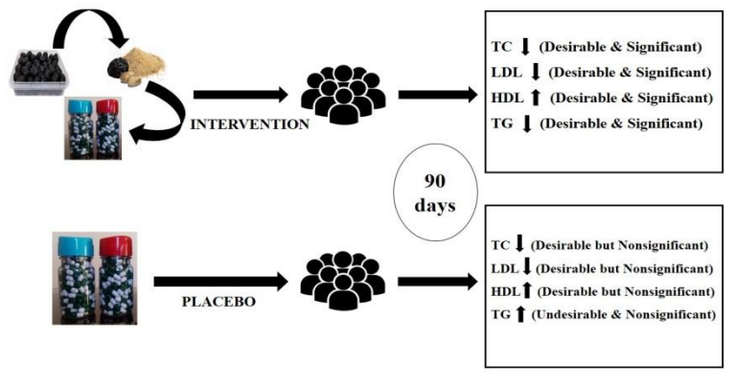
Backgrounds

Hyperlipidemia is one of the few risk factors for cardiovascular disease (CVD). Many plants are used as traditional medicines, and many of them provide comprehensive relief from CVDs. The Ajwa date contains a variety of phytochemicals that have been shown to be effective in lowering blood cholesterol. Furthermore, this date variety has a long history of use in improving cardiac function.

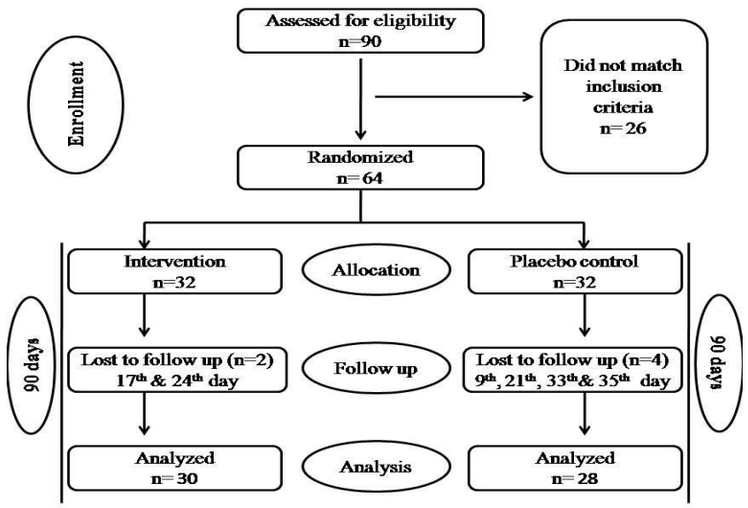
Graphical Abstract



Objectives

The purpose of this study was to determine the effects of Ajwa date seed powder on the blood cholesterol and lipid levels of a group of human volunteers.

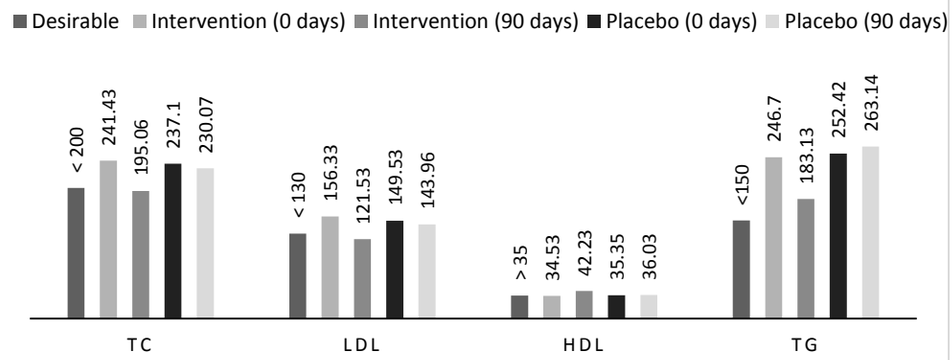
Overview of the clinical trial



Results

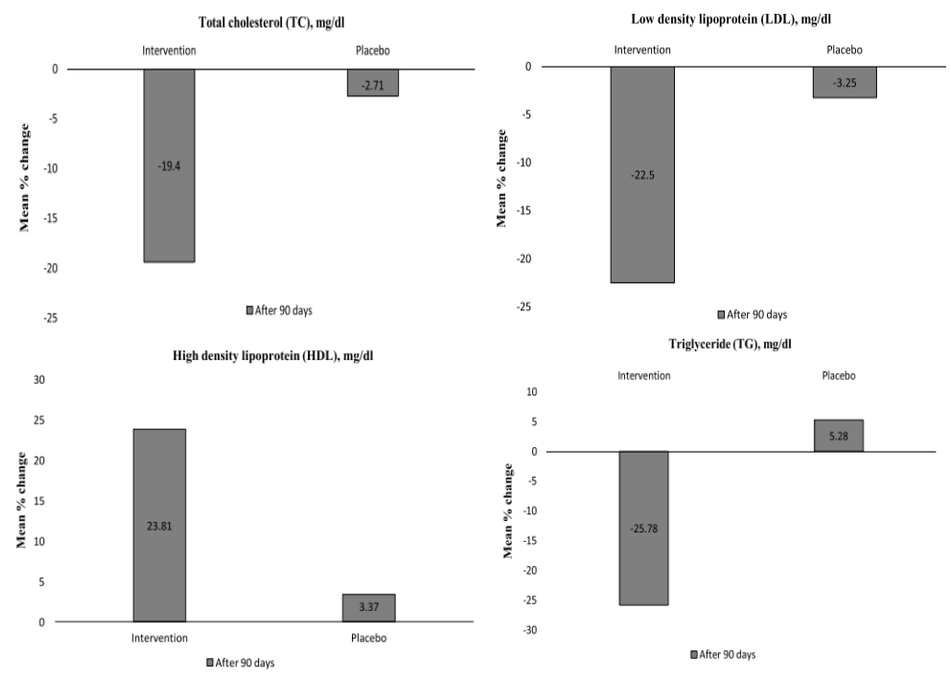
In the intervention group, the Ajwa date seed powder resulted in a 19.4%, 22.5%, and 25.78% decrease in TC, LDL, and TG levels, respectively, as well as a 23.81% rise in HDL levels, all of which were significantly better than the placebo group. The date seed showed good tolerability in human volunteers. Furthermore, after the 90-days trial, the lipid profiles of the intervention group subjects were found to be satisfactory when compared to the typical range.

Changes of lipid profile in relation to desirable limit



TC=Total cholesterol (mg/dl); LDL=Low density lipoprotein (mg/dl); HDL= High density lipoprotein (mg/dl); TG= Triglycerides (mg/dl)

% changes in TC, LDL, HDL, and TG



TC= Total cholesterol; LDL= Low-density lipoprotein; HDL= High-density lipoprotein; TG= Triglycerides

Conclusion

Finally, it is possible that Ajwa date seed powder can be used as a supplement to treat hypercholesterolemia in humans. More prospective studies with a larger subject population are required to assess the long-term hypolipidemic benefits of Ajwa date seed.

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